Cosaliency Detection and Region-of-Interest Extraction via Manifold Ranking and MRF in Remote Sensing Images

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***Abstract*— Saliency-based region-of-interest (ROI) extraction is significant for the interpretation of remote sensing images (RSIs). Recently, cosaliency detection has shown its superiority of better extraction of common ROIs by using both intraimage and interimage cues. However, most existing methods still suffer from the complex backgrounds of RSIs, resulting in incomplete ROI extraction, many false positives, and blurred boundaries. In this article, we propose a cosaliency detection framework via manifold ranking and the Markov random field (MRF) for RSIs to address these problems. First, we design a two-stage manifold ranking schema for converting single-image saliency maps (SISMs) to multi-image saliency maps (MISMs). This step takes full advantage of the correlation between images to improve the integrity of ROIs and reduce false positives. Second, we locally fuse saliency proposals by minimizing the energy function in an MRF. The design of the energy function comprehensively considers the global and local performance of saliency proposals to assign appropriate fusion weights. Finally, we generate the ROI masks by thresholding the cosaliency maps. Our approach is evaluated on four RSI datasets and compared to the state-of-the-art methods. Experimental results demonstrate the effectiveness of our model in both cosaliency detection and ROI extraction.**

***Index Terms*— Manifold ranking, Markov random field (MRF), region-of-interest (ROI) extraction, remote sensing, saliency detection.**

I. Introduction

In RECENT years, with the development of satellite and sensor technology, we can easily access a large number of remote sensing images (RSIs). How to extract useful information from massive satellite data efficiently and automatically is a fundamental and active topic in current research. Region-of-interest (ROI) extraction is an important part of RSI interpretation and can be widely used in the industry, agriculture, and military fields. In different application scenarios, ROIs usually refer to various objects, such as residential areas, airports, or oil tanks.

Saliency-based visual attention methods for extracting ROIs have gradually gained attention in recent years [1]–[8]. At the same time, considering the difference in the characteristics of RSIs and natural scene images, many saliency detection methods for remote sensing imagery have also been proposed [9]–[11]. These saliency detection models are all based on a single image. However, a group of images usually contains more abundant information than a single image. In an image group, repetitive patterns usually attract more visual attention. Hence, the concept of cosaliency detection is proposed as a branch of visual saliency detection. Cosaliency detection focus on two aspects.

1) Highlight foreground regions with respect to their surroundings in a single image.
2) Discover recurring foreground regions in a group of images.

Many cosaliency detection methods based on these two rules are proposed to extract ROIs, including bottom-up methods [16]–[21], fusion-based methods [22]–[26], and learning-based methods [27]–[31]. Bottom-up methods are widely used because of their high computational efficiency, but the detection accuracy is not satisfactory. Subsequently, many methods enhance their performance by fusing a set of complementary saliency proposals. These approaches make good use of the advantages of various saliency proposals to achieve more satisfactory results. Although learning-based methods obtain excellent results by taking prior information from training examples, they require much labor to label the data and suffer from poor generalization capability.

The previous methods proposed for natural scene images do not perform well in RSIs in three aspects: 1) ROIs cannot be recognized accurately in complex scenes with multiple salient regions; 2) background elements, such as roads and ridges, are easily misidentified; and 3) the boundaries of ROIs are blurred. This is because the visual characteristics of RSIs are very different from natural scenes. Natural scene images usually have only one or two salient objects and are located in the center of an image, while RSIs may contain multiple objects scattered throughout an image. Moreover, the background of remotely sensed images is more complex. Areas such as ridges and roads may exhibit exceptional brightness or texture features, which often leads to misjudgment.

To address these problems, we propose to use a two-stage manifold ranking to explore the correlation between images, thereby improving the integrity of salient regions and reducing false positives. To further improve the cosaliency detection

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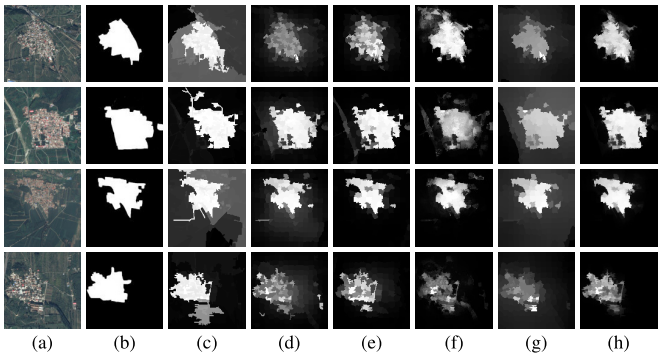


Fig. 1. Example of our fusion-based cosaliency detection. (a) Input images. (b) Ground truth. (c)–(g) Single-image saliency proposals by HS [6], MC [12], GMR [13], DSR [14], and GR [15]. (h) Fusion results by our method.

accuracy, we locally fuse multiple saliency proposals based on a Markov random field (MRF). Fig. 1 shows an example of our fusion-based method. First, we compute several single-image saliency proposals by existing methods, as shown in Fig. 1(c)–(g). Different from the work [25] that uses three cosaliency detection methods to generate cosaliency proposals, all of our proposals are generated by single-image saliency detection methods, which greatly reduces the computation complexity of the model. Second, we convert single-image saliency maps (SISMs) to multi-image saliency maps (MISMs) by the two-stage manifold ranking schema. The first-stage ranking aims to recover the salient regions that were missed by a single-image saliency proposal, and the second-stage ranking aims to eliminate no-repetitive regions. Next, by optimizing an energy function in an MRF, we accomplish the local-adaptive fusion and obtain cosaliency maps, as shown in Fig. 1(h). Finally, the binary ROI masks that represent the location of the ROIs are generated by Otsu's threshold method [32].

Different from existing MRF-based methods [24], [33], either only the global cues or local cues are considered, we appraise local and global performance simultaneously and build a two-scale energy function. Besides, although there are some cosaliency detection methods based on manifold ranking, ours is different. These existing methods directly treat the ranking results as saliency maps, while we utilize ranking results to calculate a repetitiveness map and then combine the repetitiveness map and original saliency map to obtain MISM. Not only the proposed approach uses the correlation information between the images, but also it maintains the information of the original saliency proposal.

The main contributions of the article are listed as follows.

1) We propose a two-stage manifold ranking-based method for computing repetitiveness maps, which is the key to converting SISMs to MISMs. Such a step makes full use of the interimage cues to eliminate no-repetitive regions and helps to reduce false positives.
2) We propose a saliency map fusion method by modeling a two-scale MRF, utilizing both local and global cues of each saliency map. This fusion strategy adapts the weights automatically according to the performance of each superpixel and the overall performance of the saliency map.
3) We propose an unsupervised cosaliency detection and ROI extraction framework for RSIs. The design of this framework fully considers the characteristics of RSIs. Experimental results show that the method performs well in both cosaliency detection and ROI extraction tasks.

II. Related Works

In this section, we briefly review relevant works, including single-image and multi-image saliency detection and ROI extraction models for RSIs.

A. Single-Image Saliency Detection

Saliency detection aims at capturing the uniqueness, distinctiveness, or rarity in an image. There are a lot of single-image saliency detection methods following the pioneering work of Itti *et al.* [34], which computes center-surround contrast across multiscale features to simulate human eye visual attention mechanism for saliency detection.

Jiang *et al.* [35] formulated saliency detection as a regression problem, learning a random forest regressor that maps the regional features to a saliency score. Yan *et al.* [6] proposed a hierarchical saliency (HS) model in which a multilayer method was used to analyze the saliency clues. Jiang *et al.* [12] introduced the absorbing Markov chains (MCs) into saliency detection. The saliency of each superpixel is defined as the expected time to absorption for each transient node. Yang *et al.* [13] computed the saliency via a two-stage graph-based manifold ranking (GMR). In the first stage, the similarities of image elements are ranked with background cues. In the second stage, the saliency of each element is defined as the ranking result with the initial foreground as queries. Li *et al.* [14] treated dense and sparse reconstruction (DSR) errors as an indication of saliency. Then, these two types of reconstruction errors of each pixel are integrated by Bayes' formula to form the final saliency map. Yang *et al.* [15] presented a graph-regularized (GR) saliency detection method by exploiting contrast, center, and smoothness priors. First, an initial saliency map is computed by contrast and center priors. Then, the initial saliency map is refined by minimizing an energy function, which is inspired by the smoothness prior.

These approaches work well when dealing with images with single saliency objects. However, they may fail when meeting the complex scenes.

B. Multi-Image Saliency Detection

The multi-image saliency detection methods of natural scenery are grouped into three categories in the review [36], such as bottom-up methods, fusion-based methods, and learning-based methods.

1) Bottom-Up Methods: The common feature of bottom-up cosaliency detection models is the strong dependence on manual designed cues, including intraimage and interimage cues [16]–[21].

Fu *et al.* [16] designed a two-layer cluster-based cosaliency detection (CBCS) algorithm, where one layer clusters the

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pixels on a single image, and the other layer groups the pixels on the image set. The saliency value of each cluster is then calculated from the cues of contrast, spatial, and corresponding. Finally, final saliency maps are generated by computing pixel-level saliency values based on cluster-level cues. Li *et al.* [37] proposed an efficient saliency-model-guided (ESMG) cosaliency detection framework. Given SISMs generated by an existing method, a two-stage manifold ranking scheme is adopted to obtain the guided saliency maps. Then, the guided saliency maps are fused by combining the advantages of averaging and multiplication.

2) Fusion-Based Methods: The fusion-based models make full use of the saliency maps generated by the existing saliency detection models and mine their intrinsic connections in order to assign the fusion weights. Fusion-based methods can be generally divided into three categories: fixed-weight fusion [17], [38], adaptive-weight fusion [22], [23], and locally adaptive-weight fusion [24], [25].

The fixed-weight methods use the same fusion weights for different input image groups. Since the performance of each saliency map depends on the input images, adaptive-weight methods automatically adjust the weights according to different inputs. Cao *et al.* [22] adopted a rank constraint method to obtain self-adaptively fusion weights. First, several existing (co)saliency detection methods [5], [13], [16], [39] are adopted to extract submaps for the given image group. Then, weights for fusion are obtained by the consistency among these submaps, which is mined by solving a low-rank discovery problem.

Furthermore, locally adaptive-weight fusion methods dynamically assign fusion weights according to the performance of each superpixel of different input images. Tsai *et al.* [24] improved the imagewise fusion to the region-wise fusion and proposed a cosaliency detection method by locally adaptive-weight fusion (CSLA). Taking intraimage and interimage cues and spatial consistency into account, the CSLA accomplished saliency map fusion by minimizing an energy function over a graph. Subsequently, Tsai *et al.* [25] improved the CSLA by introducing cosegmentation into cosaliency detection and developed a unified method for cosaliency detection and cosegmentation.

3) Learning-Based Methods: The learning-based models aim at automatically learning features of cosalient regions in the image group [27]–[31].

Du and Chen [27] proposed a cosaliency detection framework for a large image set. First, the image set is divided into overlapped groups, and each image in a group is divided into uniform patches. Then, they employed unsupervised random forests to express the similarity between image patches. The saliency of each patch is computed by its contrast to other patches. Intersaliency map of an image is produced by taking all groups containing the image into account. Finally, the cosaliency map is obtained by fusing a single saliency map and an intersaliency map of each image. Zhang *et al.* [28] designed a self-paced multiple-instance learning (SP-MIL) model for cosaliency detection. The MIL component of the framework aims at detecting cosaliency automatically by measuring the interimage consistency and intraimage contrast, while the SP learning component provides a sample selection mechanism, which makes the learning process more stable and robust.

C. ROI Extraction for RSIs

In recent years, some researchers have introduced saliency detection into ROI extraction in RSIs, such as the extraction of buildings [40], [41], oil tanks [42], residential areas [43], [44], aircraft [45], [46], and airports [47]–[49].

At present, most of the ROI extraction methods for RSIs are based on single-image saliency detection. Gao *et al.* [50] proposed a ship detection model based on hierarchical attention by using the relevant results of visual attention in perceptual psychology. Li and Itti [9] proposed a chip-based method that uses the improved Itti's model to generate the saliency-gist features for target detection and classification in high-resolution satellite images. Zhang and Yang [11] introduced a quaternion Fourier transform-based saliency analysis strategy and a Gaussian Pyramid-based adaptive threshold segmentation method for ROI extraction. Han *et al.* [51] proposed a weakly supervised object detection method that integrates saliency, intraclass compactness, and interclass separability in a Bayesian framework. Subsequently, Liu *et al.* [42] proposed a saliency model based on color MC for oil tank detection. Zhang *et al.* [52] proposed a self-adaptive multifeature fusion (SAMF) method based on low-rank matrix recovery. Recently, some deep learning-based saliency detection and ROI extraction methods for RSIs [44], [53] have also been proposed.

Although single-image saliency detection methods have shown their effectiveness in ROI extraction of RSIs, they ignore the relevance cues between multiple images. To address this problem, Chen and Zhang [54] proposed a joint multi-image saliency (JMS) detection algorithm to extract ROIs in a group of multispectral RSIs. The JMS algorithm first clusters all pixels on the entire image group and then calculates the saliency of each cluster by aggregating global color and shape contrast. Finally, an edge-preserving segmentation strategy is adopted to binarize the saliency maps to generate ROI masks.

III. Proposed Method

Fig. 2 shows the flowchart of our method, consisting of four steps. First, various saliency detection methods are employed as proposals for each image. Then, we perform a method based on manifold ranking to convert SISMs to MISMs. Next, cosaliency detection results are generated by fusing all the MISMs with adaptive weights decided by an MRF. Finally, we obtain binary ROI masks by applying a threshold segmentation method.

In this section, we have four points to discuss: 1) problem definition; 2) superpixel and feature extraction; 3) conversion from SISM to MISM; and 4) MRF for fusion.

A. Problem Definition

Given N input images $\{I_i\}_{i=1}^{N}$, we employ M single-image saliency detection methods to compute $N \times M$ saliency maps

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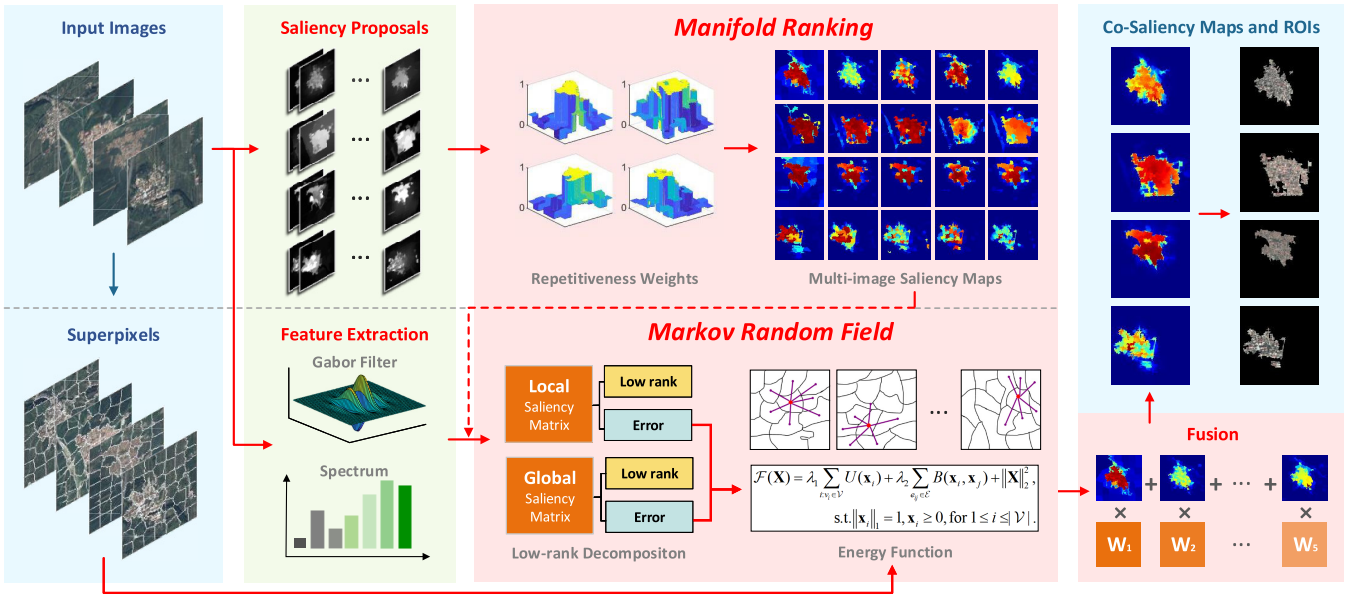


Fig. 2. Framework of the proposed method.

$\{S_j^i : 1 \le i \le N, 1 \le j \le M\}$. Each image I_i is decomposed into N_i superpixels, and the superpixel set is denoted by $\mathcal{P}^i = \{P_1^i, P_2^i, \ldots, P_{N_i}^i\}$. Total number of superpixels for all images is $N_s = \sum_{i=1}^{N} N_i$.

For converting SISMs to MISMs, our goal is to seek a matrix R^i to describe the repetitiveness of each pixel in I_i compared to salient regions of other images $\{I_m : m \neq i, 1 \le m \le N\}$, where the size of R^i is the same as I_i. Then, the MISM is $\tilde{S}_j^i := S_j^i \otimes R^i, 1 \le j \le M$, where "$\otimes$" denotes the elementwise multiplication of matrix.

For fusion, we aim to seek a plausible weight vector $\mathbf{x}_k = [x_{k,1}, x_{k,2}, \ldots, x_{k,M}]^{\mathsf{T}} \in [0,1]^M$ for each superpixel $k \in \{1, 2, \ldots, N_s\}$ and accomplish the cosaliency detection task by fusing M MISMs $\{\tilde{S}_j^i\}_{j=1}^M$. Explicitly, for a pixel $p \in P_k^i$, the fused cosaliency map CS_i can be calculated by

$$\mathrm{CS}_i(p) = \sum_{j=1}^{M} x_{k,j} \tilde{S}_j^i(p). \tag{1}$$

B. Superpixel and Feature Extraction

1) Superpixel Extraction: For each image I_i in the given image group $\{I_i\}_{i=1}^N$, we use the SLIC [55] to segment the image into N_i superpixels. Of course, other efficient superpixel segmentation methods, such as [56], can also be used. In our experiments, we set the number of superpixels in the SLIC algorithm to 50, which is discussed in Section V-C.

2) Feature Extraction: We use two types of features to describe the image, i.e., spectral and texture. On the one hand, spectral features contain the color information of RSIs, and the color is a widely used and robust feature descriptor for distinguishing various objects. On the other hand, considering that the texture features of RSIs are rich and complex, we adopt texture features extracted by Gabor filters to describe the image more accurately.

For a multispectral image with c channels, the spectral feature of each pixel is presented by a c-dimensional vector. We use the bag-of-word model to describe the spectral feature of each superpixel. All pixels of the image set are grouped into 50 clusters by using k-means [57] algorithm in the spectral space. By counting the number of pixels of each cluster contained in the ith superpixel, a 50-D histogram $\mathbf{h}_i^s$ is obtained. To be clear, the κth element of the histogram $\mathbf{h}_i^s$ indicates how many pixels in the ith superpixel belong to the κth cluster.

For texture features, we apply a set of Gabor filters to the panchromatic image. This set of filters consists of three scales of $\{2, 2\sqrt{3}, 4\}$, eight orientations $\{0, \pi/8, 2\pi/8, \ldots, 7\pi/8\}$, and two-phase offsets of $\{0, -\pi/2\}$. The settings are borrowed from [24], and there are $3 \times 8 \times 2 = 48$ combinations of these filter parameters. Hence, the texture of a pixel is represented by a 48-D vector. Similar to the spectral feature, the texture feature of a superpixel is represented by a 50-D histogram $\mathbf{h}_i^g$. We concatenate the spectral and texture features and get a 100-D descriptor for superpixel i, $\mathbf{h}_i = [\mathbf{h}_i^s; \mathbf{h}_i^g]$.

C. Conversion From SISM to MISM

We now fix the jth single-image saliency detection method and then convert SISMs $\{S_j^i, 1 \le i \le N\}$ to MISMs $\{\tilde{S}_j^i, 1 \le i \le N\}$ by taking cues among images in the group into account. Different from SISM, the saliency maps are generated by each image independently, MISM considers the correspondence between images. MISM aims to find objects not only salient in one image but also common in the group of images. Therefore, we define the interimage cue as repetitiveness; then, MISM is computed as follows:

$$\mathrm{MISM} = \mathrm{SISM} \times \mathrm{Repetitiveness}. \tag{2}$$

We next focus on calculating the repetitiveness of an image by a two-stage manifold ranking.

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1) Efficient Manifold Ranking (EMR): Zhou *et al.* [58] proposed a method for ranking data via exploiting the intrinsic manifold structure of data. In particular, we can use it to rank similarities between image pixels. Given a set of data $Y = \{y_1, y_2, \ldots, y_n\} \in \mathbb{R}^{m\times n}$. Some points in the dataset are labeled as query points, and the others will be ranked according to how similar they are to the query points. Let $\mathbf{q} = [q_1, q_2, \ldots, q_n]^{\mathsf{T}}$ denote the query vector, which means that y_i is a query if $q_i = 1$, and $q_i = 0$ otherwise. Then, we define a graph on the dataset, and $\mathbf{W} = (w_{ij})_{n\times n}$ denotes the affinity matrix. Define degree matrix $\mathbf{D}$ as a diagonal matrix where $\mathbf{D}_{ii} = \sum_{j=1}^{n} w_{ij}$. The ranking result $\mathbf{f} = [f_1, f_2, \ldots, f_n]^{\mathsf{T}}$ is obtained by minimizing an objective function

$$\mathcal{O}(\mathbf{f}) = \frac{1}{2}\left(\sum_{i,j=1}^{n} w_{ij}\left\|\frac{f_i}{\sqrt{\mathbf{D}_{ii}}} - \frac{f_j}{\sqrt{\mathbf{D}_{jj}}}\right\|^2 + \mu\sum_{i=1}^{n}\|f_i - q_i\|^2\right) \tag{3}$$

where μ is a parameter that adjusts the weight of the first item (smoothness constraint) and the second item (fitting constraint). Let $\partial\mathcal{O}/\partial\mathbf{f} = 0$, and we obtain the ranking result

$$\mathbf{f}^* = (\mathbf{I} - \alpha\mathbf{S})^{-1}\mathbf{q} \tag{4}$$

where $\mathbf{I}$ is an identity matrix, $\alpha = 1/(1+\mu)$, and $\mathbf{S} = \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$.

However, in many cases, the computational cost of the inverse matrix in (4) is very expensive due to the large size of matrix $\mathbf{W}$. To solve the issue, Xu *et al.* [59] introduced an anchor set $U = \{u_1, u_2, \ldots, u_d\} \in \mathbb{R}^{m\times d}$. Each data point $y_i \in Y$ connects with its $t(< d)$ nearest anchors, and a weight matrix $\mathbf{Z} = [z_{ki}]_{n\times d}$ measures the affinity between data point y_i and anchor u_k

$$z_{ki} = \frac{K\left(\frac{|y_i - u_i|}{\lambda_i}\right)}{\sum_{l=1}^{d} K\left(\frac{|y_i - u_l|}{\lambda_i}\right)} \tag{5}$$

where $K(\cdot)$ is the Epanechnikov quadratic kernel

$$K(\alpha) = \begin{cases} \frac{3}{4}\left(1 - t^2\right), & \text{if } |t| \le 1 \\ 0, & \text{else.} \end{cases} \tag{6}$$

$\lambda_i = |\mathbf{y}_i - \mathbf{u}_{[t]}|$ and $\mathbf{u}_{[t]}$ are the tth nearest anchors of $\mathbf{y}_i$. Then, the affinity matrix $\mathbf{W}$ is designed to be a low-rank form

$$\mathbf{W} = \mathbf{Z}^{\mathsf{T}}\mathbf{Z}. \tag{7}$$

Let $\mathbf{H} = \mathbf{Z}\mathbf{D}^{-1/2}$; then, $\mathbf{S} = \mathbf{H}^{\mathsf{T}}\mathbf{H}$. Hence, (4) can be rewritten as

$$\mathbf{f}^* = \left(\mathbf{I}_n - \alpha\mathbf{H}^{\mathsf{T}}\mathbf{H}\right)^{-1}\mathbf{q} = \left(\mathbf{I}_n - \mathbf{H}^{\mathsf{T}}\left(\mathbf{H}\mathbf{H}^{\mathsf{T}} - \frac{1}{\alpha}\mathbf{I}_d\right)^{-1}\mathbf{H}\right)\mathbf{q} \tag{8}$$

where $\mathbf{I}_n$ and $\mathbf{I}_d$ denote the identity matrix with size $n\times n$ and $d\times d$, respectively.

Parameter α is a tradeoff parameter, which is fixed at 0.8 in our experiments. We set $t = 5$ by following the default setting in [59]. The number of anchor points d is set to 100 for more efficient calculation. Please refer to [59] for more details on parameter selection in EMR.

2) Repetitiveness Measure: To simplify the denotations, we omit the subscript j of the method index in Section III-C2.

For input images $\{I_i\}_{i=1}^N$, we use the SISMs $\{S^i\}_{i=1}^N$ obtained by some method as a heuristic, adopting a two-stage EMR to measure the repetitiveness. Then, we obtain MISMs by combining the original SISMs and repetitiveness. This approach maximizes the utilization of original SISMs and reduces the false positives by leveraging the interimage evidence.

In the first stage, we use salient regions of all images as a query. In order to extract the salient regions, the threshold for binary segmentation is set as the mean value in each saliency map, i.e., $\theta^i = \texttt{mean}(S^i)$. Then, the binary segmentation map is reshaped to a vector $\mathbf{I}_i^{(1)}$

$$\mathbf{I}_i^{(1)}(k) = \begin{cases} 1, & \text{if } S^i(k) > \theta^i \\ 0, & \text{else.} \end{cases} \tag{9}$$

Next, the query vector is obtained by stacking all the vectors $\{\mathbf{I}_i^{(1)}\}_{i=1}^N$

$$\mathbf{q}^{(1)} = \left[\mathbf{I}_1^{(1)}; \mathbf{I}_2^{(1)}; \ldots; \mathbf{I}_N^{(1)}\right]. \tag{10}$$

Substitute $\mathbf{q}^{(1)}$ into (8), we get the ranking result, which is reshaped and normalized to generate coarse cosaliency maps $\{\text{CS}_i^{(1)}\}_{i=1}^N$.

The first stage ranking is essential because it can recover the missing salient areas in the SISMs. However, the results of the first stage ranking are not satisfying since the noncommon areas and backgrounds are also likely to be highlighted, as shown in Fig. 3(c). Therefore, we need the second stage ranking to further determine whether the salient areas of the first stage are repeated in the image group, so as to suppress the backgrounds and the noncommon areas that are salient in only one image.

In the second stage, we use each coarse cosaliency map obtained in the first stage as a query and then get a group of ranking results. For the ith image, we reshape and binarize the coarse cosaliency map $\text{CS}_i^{(1)}$ to a vector $\mathbf{I}_i^{(2)}$ similar to the early stage; then, the ith query is defined as

$$\mathbf{q}_i^{(2)} = \left[\mathbf{0}; \ldots; \mathbf{0}; \mathbf{I}_i^{(2)}; \mathbf{0}; \ldots; \mathbf{0}\right]. \tag{11}$$

Substitute $\mathbf{q}_i^{(2)}, i = 1, 2, \ldots, N$ into (8); we obtain N ranking results. Then, we get N cosaliency maps for each image, as shown in Fig. 3(e).

A pixel with high repetitiveness means that the pixel is similar to salient regions in other images. As shown in Fig. 3(e), a pixel that is common recurring in each row should have a high repetitiveness value. On the contrary, the pixel that is just salient in one image should have a low repetitiveness value. We define a function $\texttt{Count}(\cdot)$ to record how many times a pixel is salient among cosaliency maps. Finally, we define the repetitiveness for a pixel p as follows:

$$\tilde{R}^i(p) = \begin{cases} 1, & \text{if } \texttt{Count}(p) > N/2 \\ 0, & \text{else.} \end{cases} \tag{12}$$

To obtain more robust results, we average the repetitiveness values in each superpixel. For each pixel p in the superpixel

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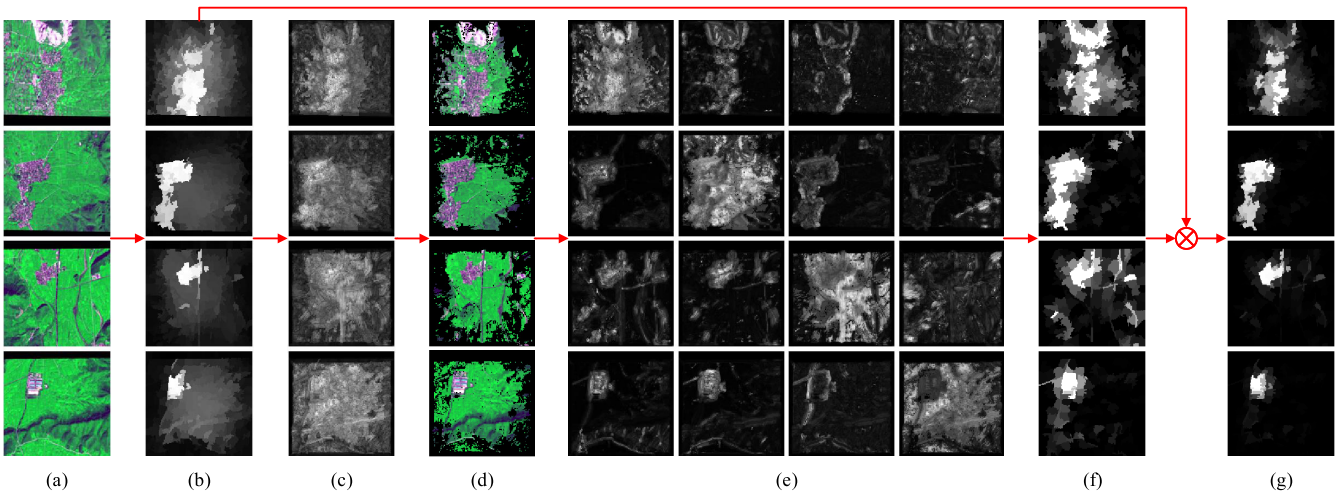


Fig. 3. Example of converting SISMs to MISMs via manifold ranking. (a) Input images. (b) Single-image saliency maps. (c) First stage ranking results. (d) Queries for the second stage ranking. (e) Second stage ranking results. (f) Repetitiveness results. (g) Multi-image saliency maps.

P_k^i, the repetitiveness is redefined as

$$R^i(p) = \texttt{mean}\left(\left\{\tilde{R}^i(q) : q \in P_k^i\right\}\right). \tag{13}$$

An example of R^i is shown in Fig. 3(f).

3) MISM Results: Finally, the MISM results are

$$\tilde{S}_j^i = S_j^i \otimes R^i \tag{14}$$

where $\otimes$ denotes the elementwise multiplication. Fig. 3(g) shows an example of MISM results. We see that MISMs are much better than the SISMs in visual effects. The results of MISMs not only retain the main salient regions but also suppress the nonrepeating regions effectively.

D. MRF for Fusion

In the previous step, we convert SISMs from several single-image saliency detection methods into MISMs. Now, we aim to find the optimal fusion weights to fuse the set of MISMs. We expect to assign adaptive fusion weights to each superpixel by following the two principles below. First, a saliency map with excellent performance should be given a higher weight. Second, two adjacent superpixels should be assigned close weights if they have similar features. Hence, MRF is adopted to complete the fusion task. In the context of MRF, we model fusion weights of each superpixel as a random element from a probability space to a measurable space. With a neighbor system given by a graph, we assume the random field satisfies Markov properties. By the Hammersley–Clifford theorem [60], the problem can be solved by minimizing an energy function on the graph.

1) Graph Construction: We first construct a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ to represent the spatial relationships among superpixels, where $\mathcal{V} = \cup_{i=1}^{N} \mathcal{V}_i$ denotes the nodes, $\mathcal{E}$ denotes edges, $\mathcal{V}_i$ consists of superpixels of the ith image, and $|\mathcal{V}| = \sum_{i=1}^{N} N_i$. Two nodes are connected by an edge $e_{ij} \in \mathcal{E}$ if their corresponding superpixels v_i and v_j are spatially adjacent. The weight of each edge is

$$s(i, j) = \exp\left(\frac{-\chi^2(\mathbf{h}_i, \mathbf{h}_j)}{\sigma}\right) \tag{15}$$

where σ is the average pairwise distance between all superpixels. Remind that $\mathbf{h}_i$ is the feature of superpixel i defined in Section III-B.

2) Energy Function: We seek plausible fusion weights $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{|\mathcal{V}|}]$ by minimizing the following energy function:

$$\mathcal{F}(\mathbf{X}) = \lambda_1 \sum_{i:v_i \in \mathcal{V}} U(\mathbf{x}_i) + \lambda_2 \sum_{e_{ij} \in \mathcal{E}} B(\mathbf{x}_i, \mathbf{x}_j) + \|\mathbf{X}\|_2^2$$
$$\text{s.t. } \|\mathbf{x}_i\|_1 = 1, \quad \mathbf{x}_i \geq \mathbf{0}, \text{ for } 1 \leq i \leq |\mathcal{V}| \tag{16}$$

where $\|\cdot\|_1$ and $\|\cdot\|_2$ denote the ℓ_1- and ℓ_2-norms, and $U(\cdot)$ and $B(\cdot, \cdot)$ denote the unary and binary terms of the energy function, respectively. The unary term implements the voting mechanism for several saliency proposals, which encourages assigning high weights to the saliency maps that perform well globally and locally. The binary term encourages the solution with desired spatial smoothness, that is, two adjacent superpixels should be assigned similar fusion weights if their features are also similar. The last term $\|\mathbf{X}\|_2^2$ is introduced for regularization, which can effectively stabilize the optimization results.

The unary term is designed via both local and global low-rank constraints. The global low-rank constraint gives an overall evaluation of the saliency map of each method, while the local low-rank constraint gives a regional assessment. Fig. 4 shows an example of calculating the unary term.

For the global low-rank constraint, our basic idea is that, if a saliency map performs poorly overall, it should be assigned a small fusion weight regardless of its local saliency value. We use a vector $\mathbf{G}_j^i$ to describe the global features of the saliency map $\tilde{S}_j^i$

$$\mathbf{G}_j^i = \left[g_{j,1}^i, g_{j,2}^i, \ldots, g_{j,N_i}^i\right] \tag{17}$$

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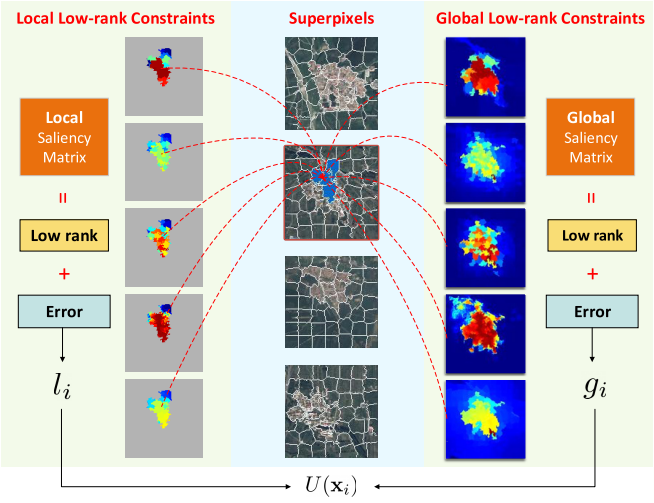


Fig. 4. Illustration of the unary term of the energy function.

where the kth element corresponds to the mean saliency value of the kth superpixel of image I_i. Stack the feature vectors of all saliency maps into a matrix

$$\mathbf{G}^i = \left[\mathbf{G}_1^i; \mathbf{G}_2^i; \ldots; \mathbf{G}_M^i\right] \in \mathbb{R}^{N_i \times M}. \tag{18}$$

We tend to assign the saliency map a better evaluation if it is consistent with others. Inspired by [61], $\mathbf{G}^i$ is decomposed into a low-rank matrix $\mathbf{L}$ and an error matrix $\mathbf{E}$. Thus, this is a low-rank matrix recovery problem defined as

$$\min_{\mathbf{L},\mathbf{E}} (\mathtt{rank}(\mathbf{L}) + \lambda \|\mathbf{E}\|_0), \text{ s.t. } \mathbf{G}^i = \mathbf{L} + \mathbf{E} \tag{19}$$

where $\|\cdot\|_0$ denotes the ℓ_0-norm and λ is a constant to control the importance of sparsity of $\mathbf{E}$. However, (19) is an NP-hard problem. Thus, RPCA [62] is adopted to solve the convex relaxation form

$$\min_{\mathbf{L},\mathbf{E}} (\|\mathbf{L}\|_* + \lambda \|\mathbf{E}\|_1), \quad \text{s.t. } \mathbf{G}^i = \mathbf{L} + \mathbf{E} \tag{20}$$

where $\|\cdot\|_*$ denotes the nuclear norm. Parameter λ is set to 0.05, which is consistent with the settings in [24], [25]. For superpixel k, we first find the image I_i in which it is in, then solve (20) for the image I_i, and denote the solution of $\mathbf{E}$ by $\mathbf{E}_k^{(g)}$.

For the local low-rank constraint, we use saliency histograms to describe local features of saliency maps. Specifically, we find n-nearest neighbors of the kth superpixel. The local features of the jth saliency map on these n superpixels are represented by a 128-D histogram $\mathbf{f}_{k,j}$. Parameter n is set to 5 in our experiments and discussed in Section V-C. Let $\mathbf{F}_k = [\mathbf{f}_{k,1}; \mathbf{f}_{k,2}; \ldots; \mathbf{f}_{k,M}]$. Similar to (19), we establish another low-rank recovery problem as

$$\min_{\mathbf{L},\mathbf{E}} (\|\mathbf{L}\|_* + \lambda \|\mathbf{E}\|_1), \quad \text{s.t. } \mathbf{F}_k = \mathbf{L} + \mathbf{E}. \tag{21}$$

Denote the solution of $\mathbf{E}$ in (21) by $\mathbf{E}_k^{(l)}$.

After solving (20) and (21), we normalize the square of the ℓ^2-norm of the jth row of matrix $\mathbf{E}_k^{(g)}$ and $\mathbf{E}_k^{(l)}$

$$l_{k,j} = \frac{\exp\left(-\left\|\mathbf{E}_k^{(l)}(j,:)\right\|_2^2\right)}{\sum_{j=1}^{M} \exp\left(-\left\|\mathbf{E}_k^{(l)}(j,:)\right\|_2^2\right)} \tag{22}$$

and

$$g_{k,j} = \frac{\exp\left(-\left\|\mathbf{E}_k^{(g)}(j,:)\right\|_2^2\right)}{\sum_{j=1}^{M} \exp\left(-\left\|\mathbf{E}_k^{(g)}(j,:)\right\|_2^2\right)}. \tag{23}$$

For energy minimization, the penalty is defined as

$$\hat{l}_{k,j} = \frac{\exp\left(1 - l_{k,j}\right)}{\sum_{j=1}^{M} \exp\left(1 - l_{k,j}\right)} \tag{24}$$

and

$$\hat{g}_{k,j} = \frac{\exp\left(1 - g_{k,j}\right)}{\sum_{j=1}^{M} \exp\left(1 - g_{k,j}\right)}. \tag{25}$$

Then, the global and local penalties are combined by letting $w_{k,j} = \hat{l}_{k,j} + \hat{g}_{k,j}$. Taking all superpixels into account, we construct the unary term of the energy function as

$$\sum_{i:v_i \in \mathcal{V}} U(\mathbf{x}_i) = \sum_{i=1}^{|\mathcal{V}|} \mathbf{w}_i^{\mathsf{T}} \mathbf{x}_i \tag{26}$$

where $\mathbf{w}_i = [w_{i,1}, w_{i,2}, \ldots, w_{i,M}]$.

The binary term is designed for smoothness constraint. For two adjacent superpixels, the greater the weight of the edge between them, the closer their fusion weights should be. Hence, the binary term of the energy function is defined as

$$\sum_{e_{ij} \in \mathcal{E}} B\left(\mathbf{x}_i, \mathbf{x}_j\right) = \sum_{e_{ij} \in \mathcal{E}} s(i,j) \|\mathbf{x}_i - \mathbf{x}_j\|_2^2. \tag{27}$$

3) Optimization: We get the final objective function by substituting (26) and (27) into (16). It is easy to see that the objective function is quadratic and, hence, has a global minimum. The optimization becomes a quadratic programming problem, which can be solved by the CVX toolbox [63] efficiently. Then, the cosaliency detection results are obtained by fusing saliency maps with the optimization result $\mathbf{X}$. Besides, we adopt a spatial refinement process [22] to further improve the performance. Finally, cosaliency maps are binarized by Otsu's algorithm [32] to produce the ROI extraction results.

We summarize the proposed method in Algorithm 1.

IV. Experimental Results

We evaluate the proposed method in this section. The benchmark datasets used for evaluation are described first. Then, evaluation metrics and implementation details are presented. Finally, the qualitative and quantitative results are reported.

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Algorithm 1 Cosaliency Detection and ROI Extraction via Manifold Ranking and MRF

Input: A group of images $\{I_i\}_{i=1}^{N}$;
for each of N input images **do**
  Generate M saliency proposals for $I_i, 1 \le i \le N$;
  Superpixel and feature extraction;
  for each of M saliency proposals **do**
    Compute repetitiveness map R^i by (13);
    Convert SISM S_j^i to MISM $\tilde{S}_j^i$ by (14);
  end for
end for
Construct MRF with energy function (16);
Optimize (16), obtain fusion weights $\{\mathbf{x}_k\}_{k=1}^{|\mathcal{V}|}$;
for each of $|\mathcal{V}|$ superpixels **do**
  Fusion with weights $\mathbf{x}_k$;
end for
ROI extraction by thresholding cosaliency maps;
Output: Cosaliency maps $\{\mathrm{CS}_i\}_{i=1}^{N}$ and ROI masks.

A. Datasets

Our experiments are all carried out on RSIs. We constructed a dataset with 112 RSIs from two satellites and Google Earth. The GeoEye-1 satellite provides imagery at 0.41-m panchromatic and 1.65-m multispectral (blue, green, red, and near-infrared) resolutions. The SPOT-5 satellite provides imagery at 2.5-m panchromatic and 10-m multispectral (green, red, near-infrared, and short-wave infrared) resolutions. The GeoEye-1 and SPOT-5 datasets contain eight and 20 RSIs, respectively, and their common ROIs are residential areas. The Google Earth is an optical RSI dataset with a spatial resolution of 0.5–1 m and contains three bands of red, green, and blue. The Google Earth dataset contains 72 images with residential areas as ROIs and 12 images with airports as residential areas. The size of the images in all datasets is 1024×1024. Because of the differences in the resolution, spectrum, and ROI of images from different sources, each type of ROI from different sources independently makes a group.

B. Evaluation Criteria

Given a threshold, a saliency map can be binarized into ROIs and background. Then, we evaluate the saliency map by comparing the ROI masks and the ground-truth segmentation. The precision ($\mathcal{P}$) and recall ($\mathcal{R}$) are defined by

$$\mathcal{P} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad \mathcal{R} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{28}$$

where TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively.

Since precision and recall are a pair of contradictory measures, it is essential to evaluate them simultaneously. The threshold is taken from 256 levels between 0 and 1 to draw the precision–recall (PR) curve and the receiver operating characteristic (ROC) curve. Let AP and AUC represent the area below the PR curve and the AUC curve, respectively. Also, with Otsu's adaptive threshold, F-measure is computed by the harmonic mean of precision and recall

$$F_\beta = \frac{\left(1+\beta^2\right) \times \mathcal{P} \times \mathcal{R}}{\beta^2 \times \mathcal{P} + \mathcal{R}}. \tag{29}$$

Parameter β is used to adjust the weight of precision and recall. In this article, we set $\beta^2 = 0.3$ to emphasize recall more, which is used in [64].

As introduced in [65], the traditional evaluation criteria may suffer the interpolation, dependence, and equal importance flaws. To solve this problem, we adopt the weighted F-measure F_β^w proposed in [65]

$$F_\beta^w = \frac{\left(1+\beta^2\right) \times \mathcal{P}^w \times \mathcal{R}^w}{\beta^2 \times \mathcal{P}^w + \mathcal{R}^w} \tag{30}$$

where $\mathcal{P}^w$ and $\mathcal{R}^w$ denote weighted precision and weighted recall, respectively. In (30), we set $\beta^2 = 1$ by using the original setting in [65].

C. Implementation Details

We choose single-image saliency proposals by comprehensively considering simplicity, efficiency, and complementary effects. We refer to the benchmark [66] of salient object detection algorithms and choose five single-image saliency proposals, i.e., HS [6], MC [12], GMR [13], DSR [14], and GR [15]. HS can provide saliency proposals with clear boundaries and uniform brightness. GMR and GR measure saliency based on the contrast and center priors, so they may miss a saliency region near the boundaries of the image. However, MC performs well in recognition of the areas connected to image boundaries, so it can effectively make up for their shortcomings. Besides, DSR is a method that is good at preserving boundaries of saliency regions but does not perform well when meeting multiple saliency regions in an image. In general, the better the proposals, the better the fusion results, so we choose these popular and effective methods with complementary performance.

We set $\lambda_1 = 8$ and $\lambda_2 = 1$ in the energy function and conduct grid search experiments to ensure that the settings are appropriate. The results of grid search experiments are presented in Section V-C.

D. Comparative Results With Other Methods

We compare our proposed method with ten models, including five single-image saliency proposals and five state-of-the-art (co)saliency detection methods, i.e., SAMF [52], CBCS [16], ESMG [37], JMS [54], and CSLA [24].

SAMF is a single-image saliency detection method based on low-rank matrix recovery. CBCS is a bottom-up method that learns the corresponding between multiple images through the clustering process and uses three visual attention cues of contrast, spatial, and corresponding to measure the cluster saliency. ESMG obtains the cosaliency maps through a manifold ranking scheme. Different from ESMG, which treats the ranking results directly as cosaliency maps, we design a new method for calculating the repetitiveness and yield MISMs by combining the original saliency maps and the repetitiveness.

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TABLE I

EFFECTIVENESS EVALUATION OF COSALIENCY DETECTION RESULTS ON FOUR DATASETS. THE BEST RESULTS ARE MARKED IN **BOLD**; "-" MEANS THAT THE SCORE CANNOT BE CALCULATED CORRECTLY

Method	GoeEye-1				SPOT-5				Google Earth-Residential				Google Earth-Airport			
	AP	AUC	F_β	F_β^ω	AP	AUC	F_β	F_β^ω	AP	AUC	F_β	F_β^ω	AP	AUC	F_β	F_β^ω
HS [6]	0.671	0.884	0.737	0.540	0.372	0.772	0.336	0.283	0.709	0.875	0.623	0.452	0.423	0.831	0.263	0.261
MC [12]	0.666	0.924	0.655	0.470	0.699	0.885	0.496	0.337	0.791	0.922	0.793	0.477	0.780	0.953	0.536	0.286
GMR [13]	0.702	0.918	0.582	0.523	0.673	0.846	0.593	0.363	0.783	0.907	0.771	0.579	0.789	0.954	0.621	0.404
DSR [14]	0.548	0.898	0.654	0.481	0.699	0.915	0.714	0.467	0.767	0.946	0.780	0.543	0.648	0.918	0.619	0.377
GR [15]	-	0.834	0.625	0.312	-	0.779	0.489	0.224	-	0.801	0.728	0.372	-	0.684	0.481	0.161
SAMF [52]	0.312	0.768	0.324	0.233	0.323	0.860	0.325	0.206	0.749	**0.966**	0.721	0.332	0.608	0.936	0.538	0.200
CBCS [16]	0.723	0.932	0.643	0.351	0.510	0.880	0.318	0.213	0.761	0.954	0.711	0.338	0.607	0.831	0.523	0.224
ESMG [37]	0.728	0.924	0.787	0.586	0.635	0.934	0.644	0.493	0.807	0.925	0.831	0.614	0.735	0.923	0.697	0.446
JMS [54]	0.435	0.805	0.444	0.420	-	0.718	0.332	0.290	0.572	0.819	0.610	0.520	0.041	0.439	0.027	0.038
CSLA [24]	0.569	0.918	0.611	0.453	0.694	**0.939**	0.530	0.404	0.811	0.965	0.771	0.508	0.769	0.965	0.648	0.383
OURS	**0.840**	**0.966**	**0.840**	**0.625**	**0.830**	0.926	**0.788**	**0.526**	**0.902**	**0.966**	**0.874**	**0.643**	**0.816**	**0.969**	**0.711**	**0.471**

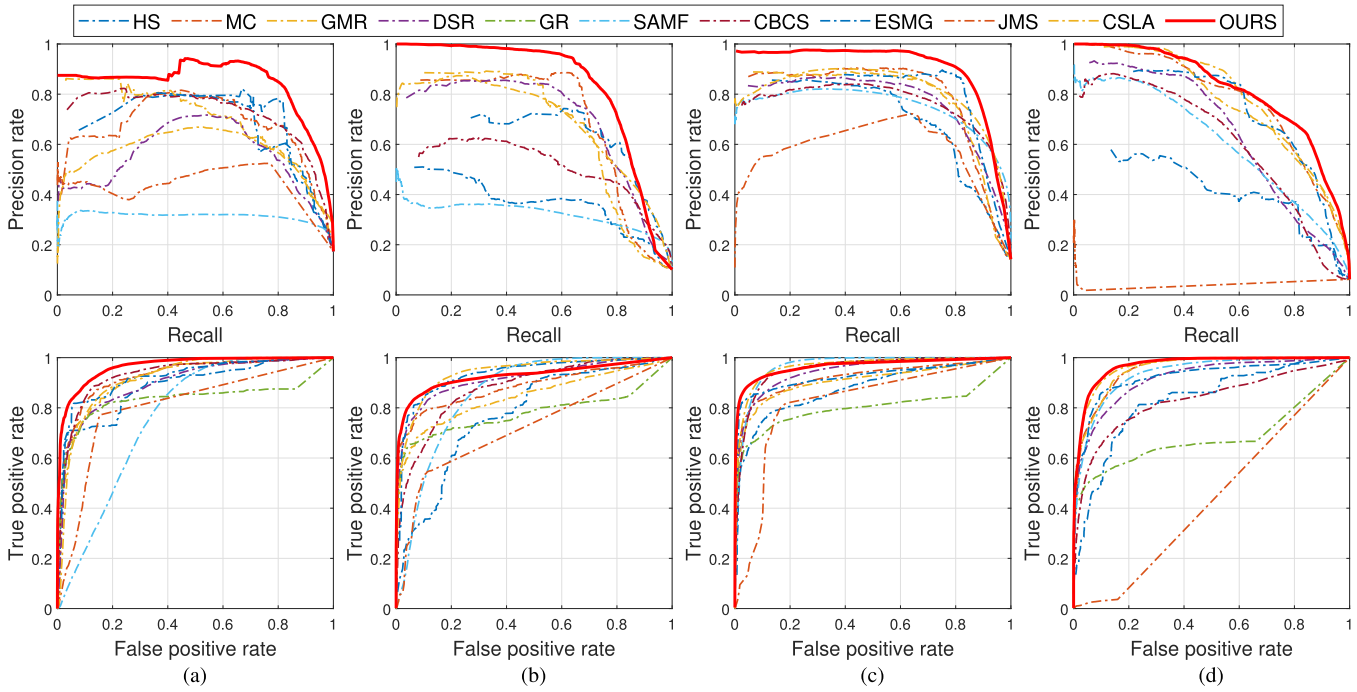


Fig. 5. Objective evaluation of PR and ROC curves on four datasets. (a) PR and ROC curves on GeoEye-1 dataset. (b) PR and ROC curves on the SPOT-5 dataset. (c) PR and ROC curves on the Google Earth-Residential dataset. (d) PR and ROC curves on the Google Earth-Airport dataset. Our proposed model is plotted in a red solid line, while other comparative models are in dash lines with different colors.

JMS is a cluster-based multi-image saliency detection method for RSIs. CSLA is a fusion-based approach that uses both SISMs and MISMs as saliency proposals. Unlike CSLA, our proposed method only adopts SISMs as saliency proposals because we specially designed a module that converts SISMs to MISMs. Besides, CSLA only considers the local features of saliency proposals, while our proposed method comprehensively considers the overall and local performance to evaluate the saliency proposals and assign fusion weights.

1) Quantitative Assessment: Table I presents the AP, AUC, F_β, F_β^ω scores of ours, and comparative methods, and Fig. 5 shows the PR and ROC curves. Compared with the five saliency proposals, all the objective metrics are greatly improved. We further compare our method with five (co)saliency detection methods, and our approach achieves better performance than all the competing approaches.

On the GeoEye-1 and Google Earth-Residential datasets, our approach achieves the best results in all metrics. On the SPOT-5 dataset, our method has slightly lower performance in AUC but has better results in AP, F_β, and F_β^ω. Compared with the datasets where the residential areas are salient objects, the objective metrics on the Google Earth-Airport dataset are less satisfactory. The reason is that the shape of the airport is more irregular than that of the residential areas, and the roads and houses in the background will also interfere with the extraction of airports. Other comparative methods also face the same challenges, and our method still achieves the best results.

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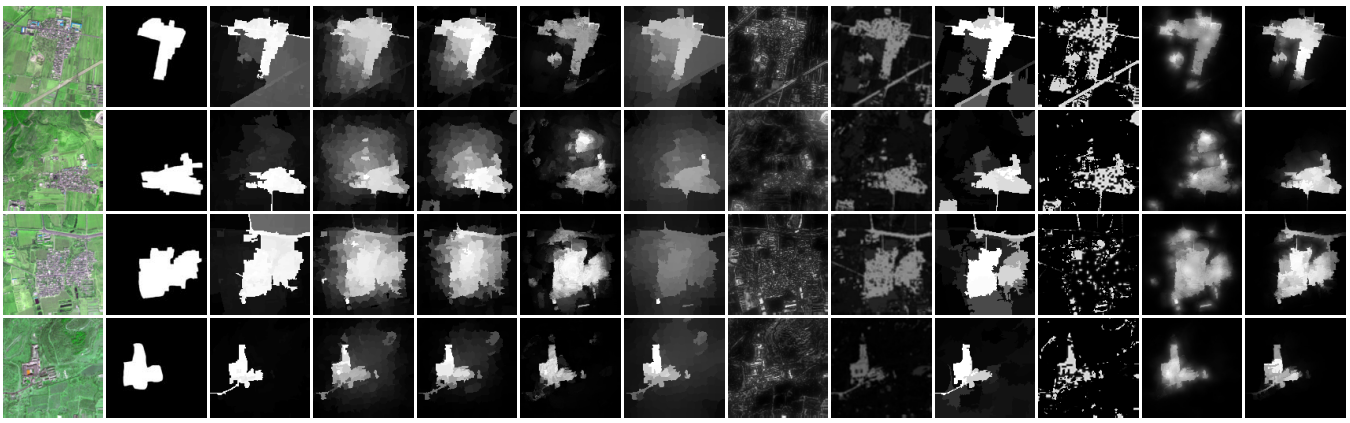


Fig. 6. Visual comparison of saliency maps for the GeoEye-1 dataset. (a) Input images. (b) Ground truth. (c)–(g) Single-image saliency proposals, including HS, MC, GMR, DSR, and GR. (h)–(l) Results by (co)saliency detection methods including SAMF, CBCS, ESMG, JMS, and CSLA. (m) Ours.

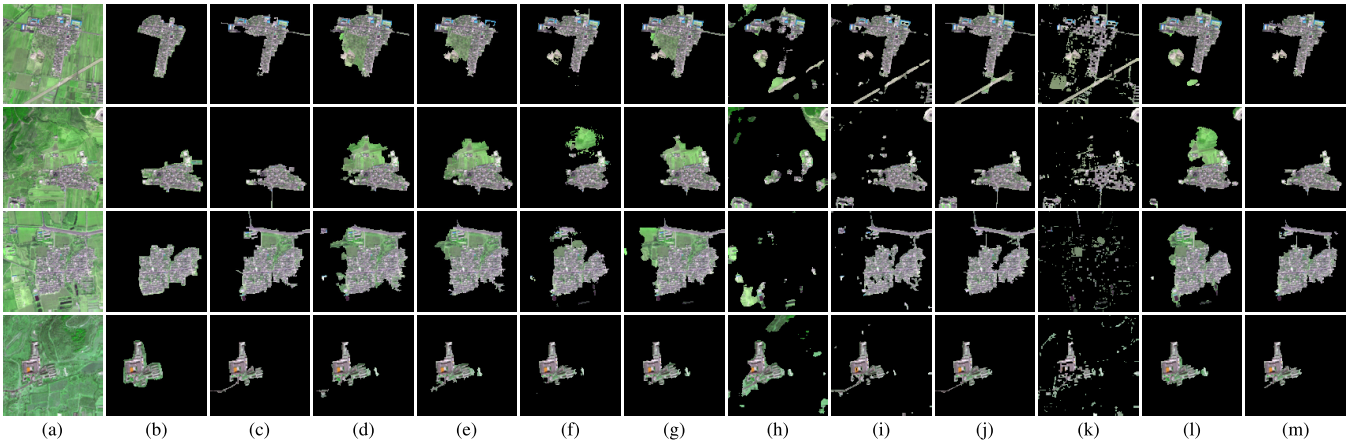


Fig. 7. Visual comparison of ROI extraction results for the GeoEye-1 dataset. (a) Input images. (b) Ground truth. (c)–(g) Single-image saliency proposals, including HS, MC, GMR, DSR, and GR. (h)–(l) Results by (co)saliency detection methods, including SAMF, CBCS, ESMG, JMS, and CSLA. (m) Ours.

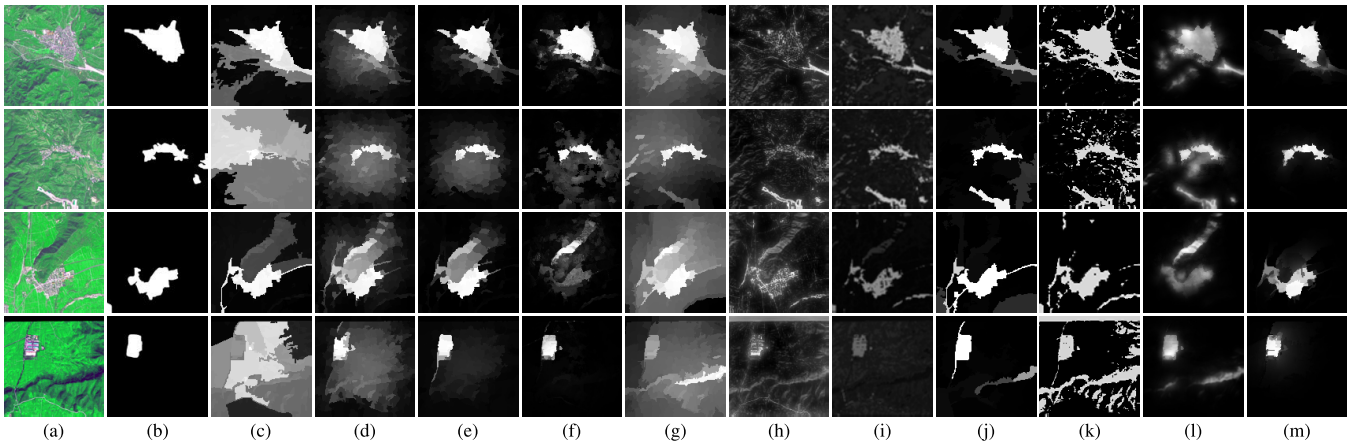


Fig. 8. Visual comparison of saliency maps for the SPOT-5 dataset. (a) Input images. (b) Ground truth. (c)–(g) Single-image saliency proposals, including HS, MC, GMR, DSR, and GR. (h)–(l) Results by (co)saliency detection methods, including SAMF, CBCS, ESMG, JMS, and CSLA. (m) Ours.

2) Qualitative Assessment: We randomly pick a group of four images from each dataset as a display, and visual comparative results of cosaliency detection and ROI extraction are shown in Figs. 6–13.

The six single-image saliency detection methods, i.e., HS, MC, GMR, DSR, GR, and SAMF, cannot present satisfactory results on most test images. Because they either misinterpret backgrounds, such as roads and ridges as salient areas or miss

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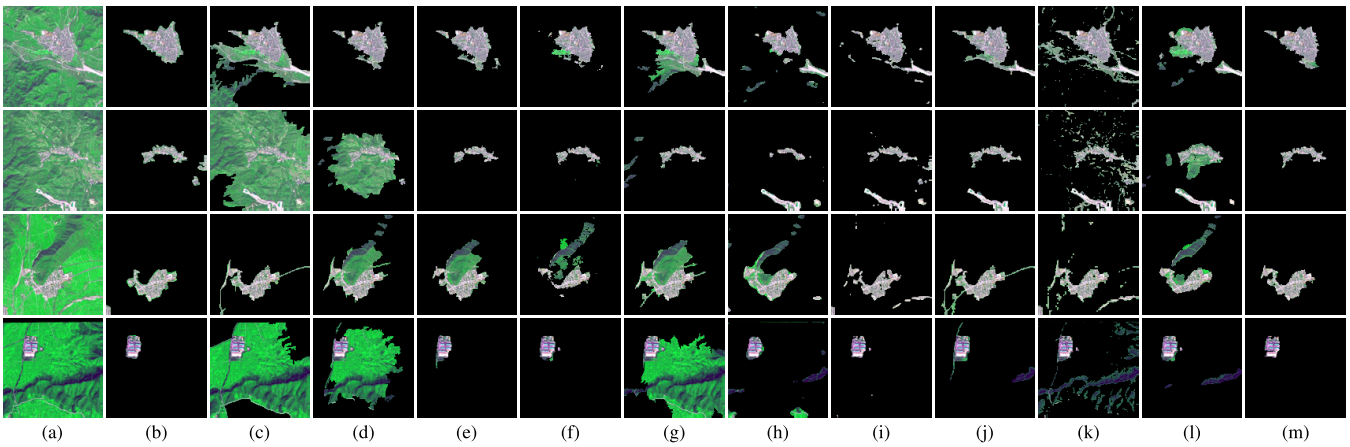


Fig. 9. Visual comparison of ROI extraction results for the SPOT-5 dataset. (a) Input images. (b) Ground truth. (c)–(g) Single-image saliency proposals, including HS, MC, GMR, DSR, and GR. (h)–(l) Results by (co)saliency detection methods, including SAMF, CBCS, ESMG, JMS, and CSLA. (m) Ours.

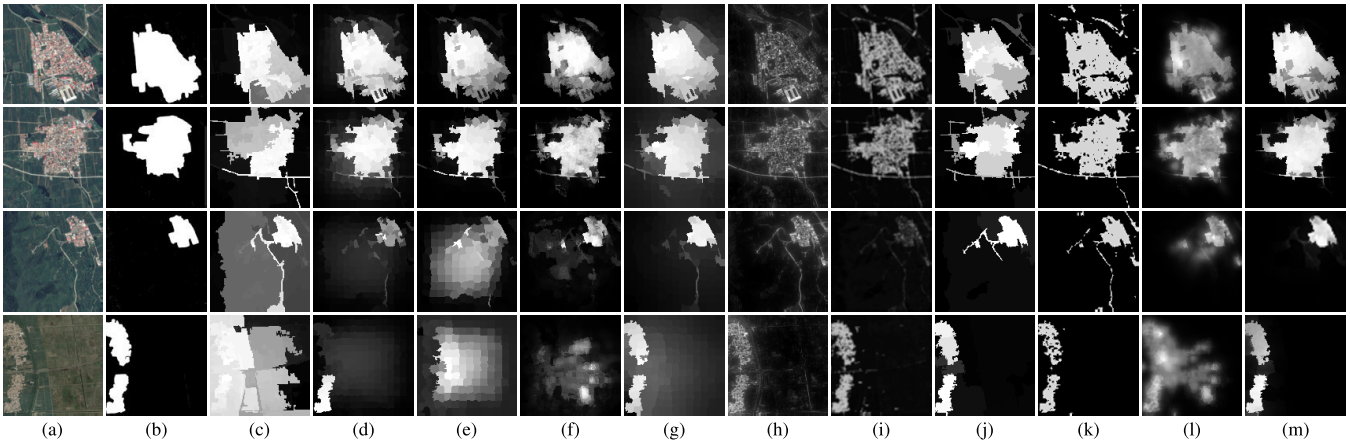


Fig. 10. Visual comparison of saliency maps for the Google Earth-Residential dataset. (a) Input images. (b) Ground truth. (c)–(g) Single-image saliency proposals, including HS, MC, GMR, DSR, and GR. (h)–(l) Results by (co)saliency detection methods, including SAMF, CBCS, ESMG, JMS, and CSLA. (m) Ours.

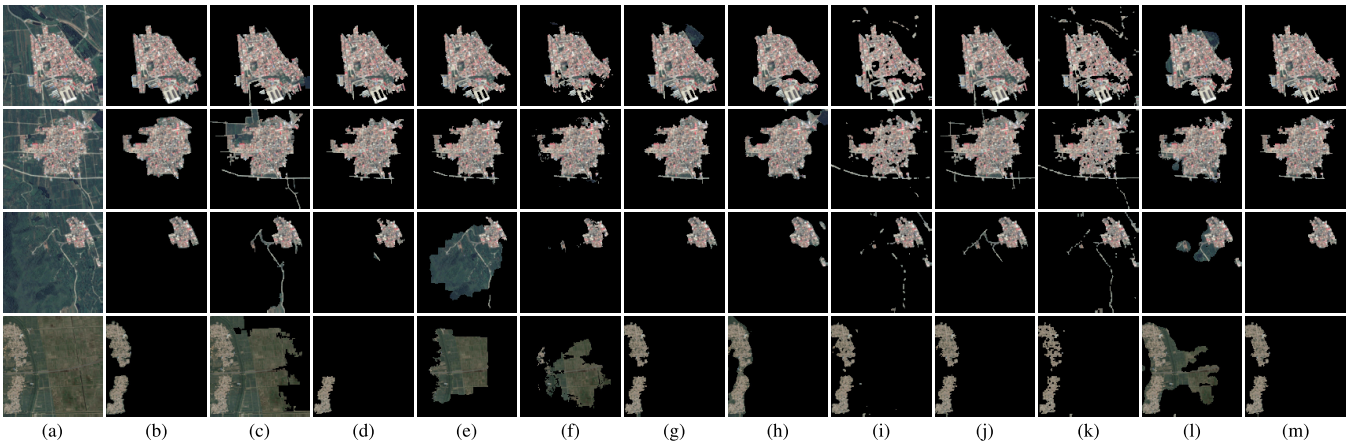


Fig. 11. Visual comparison of ROI extraction results for the Google Earth-Residential dataset. (a) Input images. (b) Ground truth. (c)–(g) Single-image saliency proposals, including HS, MC, GMR, DSR, and GR. (h)–(l) Results by (co)saliency detection methods, including SAMF, CBCS, ESMG, JMS, and CSLA. (m) Ours.

salient regions, HS has many false positives. GMR tends to assign high saliency to the center of the image, so the detection result is inaccurate when the saliency object is close to the edges, as shown in lines 3 and 4 of Fig. 10. MC, DSR, and GR algorithms may miss some of the salient regions or may not detect salient objects at all, as shown in line 4 of Fig. 10

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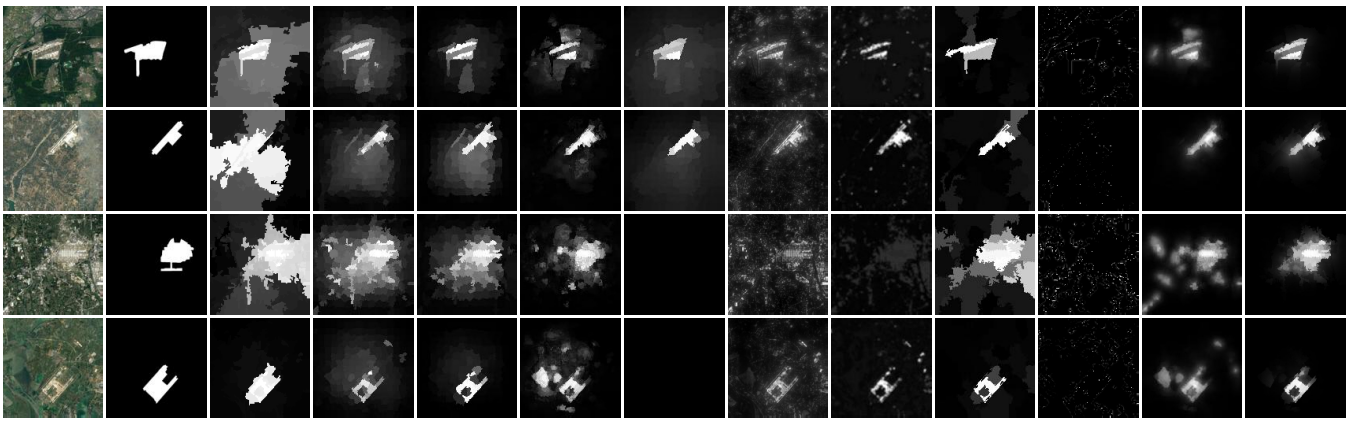


Fig. 12. Visual comparison of saliency maps for the Google Earth-Airport dataset. (a) Input images. (b) Ground truth. (c)–(g) Single-image saliency proposals, including HS, MC, GMR, DSR, and GR. (h)–(l) Results by (co)saliency detection methods, including SAMF, CBCS, ESMG, JMS, and CSLA. (m) Ours.

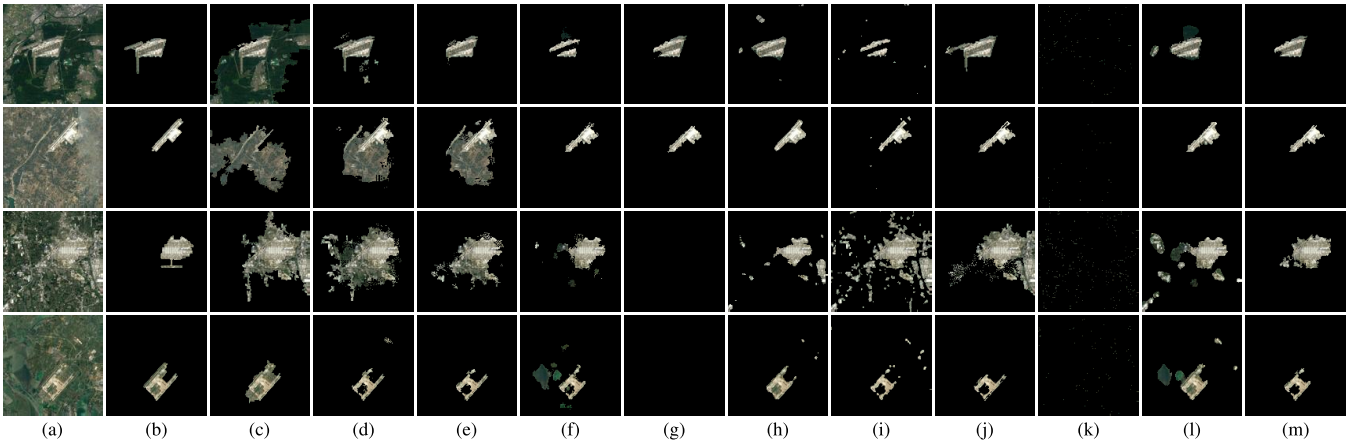


Fig. 13. Visual comparison of ROI extraction results for the Google Earth-Airport dataset. (a) Input images. (b) Ground truth. (c)–(g) Single-image saliency proposals, including HS, MC, GMR, DSR, and GR. (h)–(l) Results by (co)saliency detection methods, including SAMF, CBCS, ESMG, JMS, and CSLA. (m) Ours.

and lines 3 and 4 of Fig. 12. The brightness of the saliency maps generated by SAMF is not uniform enough and contains a lot of noise.

The four cosaliency detection methods, i.e., ESMG, CBCS, JMS, and CSLA, take interimage cues into account, which helps to suppress false positives. ESMG can identify the most salient regions but misjudges many background elements, such as roads and ridges. CBCS can detect salient regions completely but assigns a very low saliency to most of the salient regions, resulting in low contrast of the generated saliency maps. CSLA can generally identify ROIs, but the boundaries are very blurred, and there is noise in the background. JMS performs. JMS can detect residential areas well but failed completely on the airport dataset.

Our proposed method yields high-quality cosaliency maps in two aspects. First, our cosaliency maps can accurately highlight salient regions and effectively eliminate areas, such as roads and ridges, which are salient in one image but not repeated in other images. Second, our proposed method gives a clear boundary between the salient regions and the backgrounds. Therefore, the results of our method achieve the best visual effects among all comparative methods.

TABLE II
COMPARISON OF THE AVERAGE RUNNING TIME

Method	Code Type	Time (s)
MC [12]	Matlab & C++	0.0301
JMS [54]	Matlab	0.0401
GMR [13]	Matlab	0.1297
CBCS [16]	Matlab	0.1372
HS [6]	Executable	0.2314
GR [15]	Matlab & C++	0.2623
ESMG [37]	Matlab	0.5072
SAMF [52]	Matlab	0.6403
DSR [14]	Matlab	1.3648
OURS	Matlab & C++	3.5949
CSLA [24]	Matlab & C++	64.7420

3) Efficiency Evaluation: We present the time comparison of ten comparative models and our proposal in Table II. All experiments are carried out on a desktop with an

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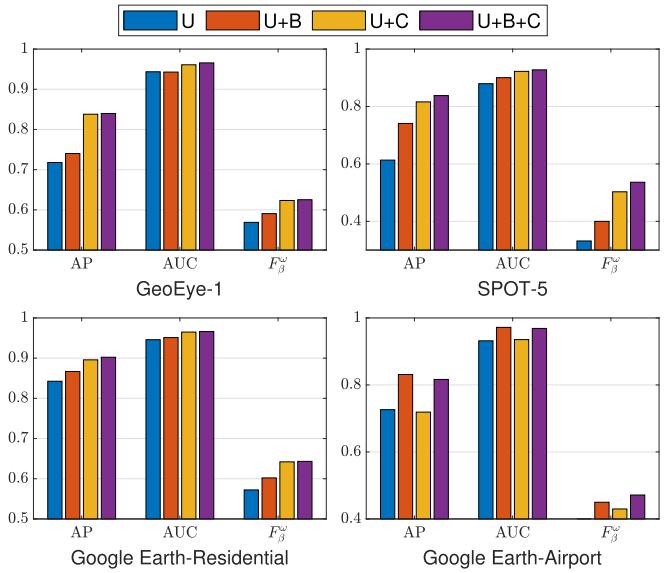


Fig. 14. Ablation studies on four datasets in AP, AUC, and F_β^ω. "U" and "B" denote unary and binary terms in energy function, respectively. "C" denotes the conversion step from SISMs to MISMs.

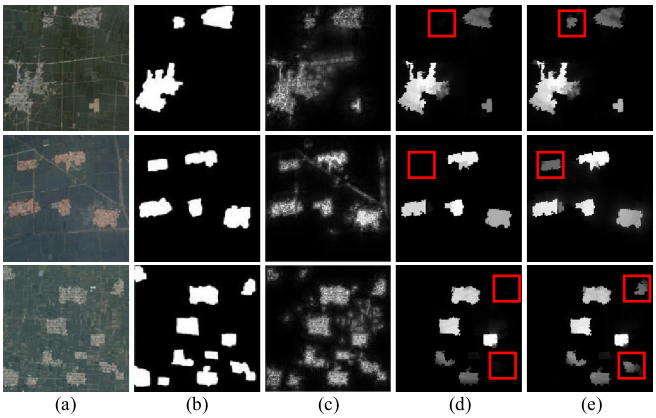


Fig. 15. Visual comparison of single-stage ranking results and two-stage ranking results. (a) Input images. (b) Ground truth. (c)–(e) Saliency detection results by using the first-stage ranking, the second-stage ranking, and the two-stage ranking, respectively.

TABLE III
ABLATION STUDIES OF SINGLE-STAGE RANKING AND TWO-STAGE RANKING. "STAGE 1" AND "STAGE 2" DENOTE THE FIRST- AND SECOND-STAGE RANKINGS, RESPECTIVELY. THE BEST PERFORMANCE IS SHOWN IN **BOLD**

Stage 1	Stage 2	GeoEye-1			SPOT-5		
		AP	AUC	F_β^ω	AP	AUC	F_β^ω
✓	×	0.696	0.919	0.385	0.566	0.891	0.280
×	✓	0.827	**0.966**	0.624	0.822	0.919	0.519
✓	✓	**0.840**	**0.966**	**0.625**	**0.830**	**0.926**	**0.526**
Stage 1	Stage 2	Google Earth-Residential			Google Earth-Airport		
		AP	AUC	F_β^ω	AP	AUC	F_β^ω
✓	×	0.801	0.956	0.444	0.367	0.877	0.177
×	✓	0.900	0.962	0.639	0.804	0.959	0.461
✓	✓	**0.902**	**0.966**	**0.643**	**0.816**	**0.969**	**0.471**

Intel i7-4790 3.60-GHz CPU and 16-GB RAM. For a fair comparison, due to the limitation of the input size of the CSLA method, the input images of all comparative methods are resized to 128×128. We evaluate all methods ten times and calculate the average running time. Because the single-image saliency detection methods can be freely changed, the reported running time of our method does not include the calculation of single-image saliency proposals.

In this article, we construct an MRF to fuse multiple saliency proposals to achieve unsupervised cosaliency detection, so we sacrificed some computational efficiency to achieve more accurate calculation results. Although our proposal is relatively slower, its results are better, which partially compensates for the speed.

V. DISCUSSION

In this section, we analyze our model in three aspects, ablation studies, parameter analysis, and feature selection. Besides, we will discuss the performance of large-scale RSIs and analyze the limitations of our proposed model.

A. Ablation Studies

We perform ablation studies on all datasets to evaluate the contributions of each component of the overall architecture. The main components include the conversion step from SISMs to MISMs and the unary and binary terms in the energy function of MRF. The first model, denoted by "U," uses only the unary term of the energy function to fuse saliency proposals. That is, the conversion step from SISMs to MISMs and the binary term in energy function are removed. The second model, denoted by "U + C," is constructed by removing the binary term in the energy function. The third model, denoted by "U + B," is constructed by removing the conversion step from SISMs to MISMs. The last one is our full model proposed in this article, denoted by "U + B + C."

As shown in Fig. 14, we find that adding components, "U," "B," and "C," step by step can progressively improve the performance. The step of converting SISMs to MISMs based on the manifold ranking scheme can significantly improve the AP score, especially in the GeoEye-1 and Google Earth-Residential datasets. Besides, adding a binary term for smoothness constraints can also improve the objective metrics of the model.

We also performed ablation experiments to evaluate the contribution of each stage ranking in the two-stage manifold ranking. We compare the proposed model with two models with single-stage manifold ranking, which use only the first-stage ranking and only the second-stage ranking, respectively. Fig. 15 shows the visual comparison of single-stage ranking results and two-stage ranking results. We find that the results of using only the first-stage ranking have many false alarms, and the visual effect is not satisfactory. The results using only the second-stage ranking may miss some salient areas, as shown in the red boxes in Fig. 15(d). We find that applying two-stage ranking can obtain better cosaliency

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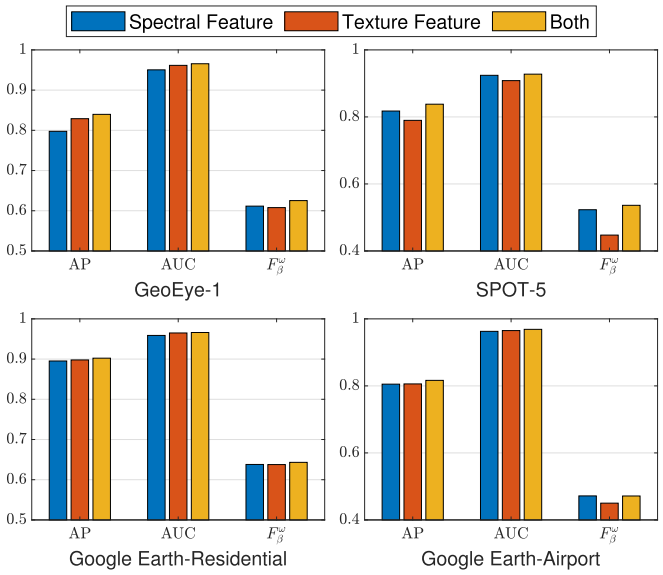


Fig. 16. Feature selection studies on four datasets in AP, AUC, and F_β^ω. The blue, red, and yellow bars show the experimental results with spectral feature, texture feature, and both of them, respectively.

detection results, as shown in Fig. 15(e). Table III shows the comparison results of three objective metrics (AP, AUC, and F_β^ω) on four datasets. We find that using only the first-stage ranking yields the worst results, and using the two-stage ranking can obtain the best results on all evaluation metrics.

B. Feature Selection

We adopt spectral and texture features to describe RSIs. To verify the effectiveness of these two types of features, we conduct comparative experiments with three models, using spectral features, texture features, and both of them, respectively. The comparative results on the four datasets are shown in Fig. 16. We reached the same conclusion on all datasets that, using both spectral and texture features, can obtain better saliency detection results.

C. Parameter Analysis

Our model has four hyperparameters, the number of superpixels, the neighborhood size n, and the weighting parameters λ_1 and λ_2 in the energy function. We discuss the selection of these parameters as follows.

As we can see in Fig. 17, the F_β^ω score continues to increase as the number of superpixels increases from 1 to 50. This is because too few superpixels will invalidate the locally adaptive fusion strategy, causing regions with large feature differences to use the same fusion weights, thus reducing the quality of fused cosaliency maps. When the number of superpixels exceeds 50, the F_β^ω score tends to stabilize. As the number of superpixels increases, the number of nodes in the MRF also increases, resulting in a significant increase in the computational complexity of objective function optimization in MRF. Thus, the appropriate number of superpixels should balance the quality of the results with computational efficiency. In this article, we set the number of superpixels to 50 according to the experimental results. It is worth noting that the exact number of superpixels is around 50, which is determined by the SLIC [55] algorithm.

TABLE IV
COMPARISON OF OBJECTIVE METRICS WITH AND WITHOUT THE IMAGE TILING STRATEGY

Image tiling	AP	AUC	F_β	F_β^ω
×	0.490	0.824	0.363	0.316
✓	0.544	0.916	0.495	0.436

Parameter n is the neighborhood size of superpixels, which is defined in the local low-rank constraints in the unary term of the energy function. To prevent the saliency histogram of a superpixel from being too sparse when the number of pixels of a superpixel is small, we choose the n nearest neighbors of a superpixel to improve the robustness of the saliency features. We investigated the effect of n on the experimental results. As shown in Fig. 17(b), the F_β^ω score is not sensitive to the parameter n because, at a setting of 50 superpixels, even a single superpixel contains enough pixels to describe the saliency features. In this article, we set n to 5 and achieve stable results on all datasets.

We choose the appropriate values of λ_1 and λ_2 based on the grid search experiments. The results of grid search experiments are shown in Fig. 17(c) and (d). As can be seen, the parameters λ_1 and λ_2 slightly influence the performance, depending on the target datasets. As λ_1 increases from 2 to 8, the objective metrics on the Google Earth-Residential and Google Earth-Airport datasets increase, while the performance on the SPOT-5 dataset decreases. As λ_1 continues to increase, objective metrics reach a stable level on the Google Earth-Residential dataset, while performance slightly declines on SPOT-5 and Google Earth-Airport datasets. The variation of λ_2 has roughly the opposite effect on the metric as λ_1 because it emphasizes the other constraint in the energy function. We find that the effect of the two weighing parameters in the SPOT-5 dataset is different from other datasets. The reason is that the spatial resolution of the SPOT-5 scenes is lower and the salient areas occupy a smaller area than the Google Earth scenes. Thus, the binary term that can increase the smoothness of the results is of higher importance. In this article, we comprehensively consider the variation of objective metric on the four datasets, setting $\lambda_1 = 8$ and $\lambda_2 = 1$.

D. Study of Large-Scale RSIs

Large size is another characteristic of RSIs, so we also investigate how the proposed method performs on large-scale RSIs. We constructed a challenging large-scale RSI dataset from Google Earth. The dataset provides imagery at 1-m multispectral resolution with a size of 8000 × 8000. We first performed ROI detection directly on these large-scale images, as shown in Fig. 18(c). Another strategy is to cut large-scale RSIs into small tiles and then perform ROI detection [67], [68]. This strategy is called image tiling. We cut each image into 49 tiles of 2000 × 2000 with 50% overlap, and the final results are obtained by splicing the results of these tiles,

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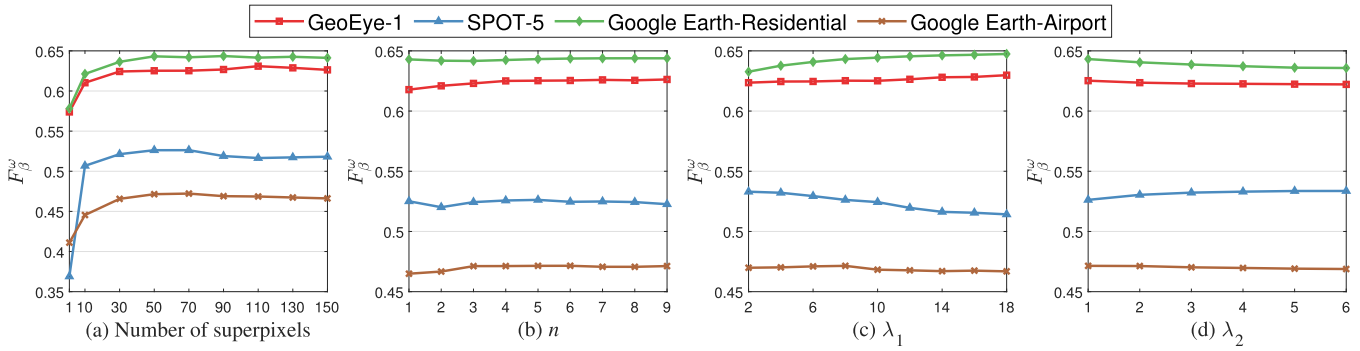


Fig. 17. Parameter analysis on the number of superpixels, n, λ_1, and λ_2. When the investigated variable changes, other variables are fixed and take default values. The default values of the number of superpixels, n, λ_1, and λ_2, are 50, 5, 8, and 1. (a) F_β^ω with the variation of the number of superpixels. (b) F_β^ω with the variation of n. (c) F_β^ω with the variation of λ_1. (d) F_β^ω with the variation of λ_2.

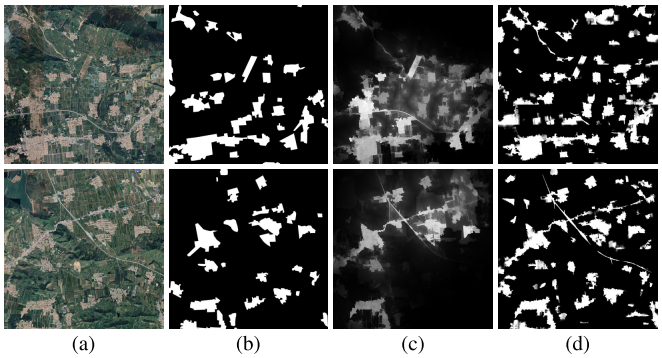


Fig. 18. Visual comparison of saliency maps for the large-scale dataset. (a) Input image group. (b) Ground truth. (c) Results of our method without image tiling. (d) Results of our method with image tiling.

as shown in Fig. 18(d). The comparison of objective metrics is shown in Table IV.

We find that the results of direct detection are not satisfactory, and the image tiling strategy can significantly improve performance. This observation is mainly because saliency characterizes the most prominent parts of the scene to the observer globally. In large-scale images, ROIs may be too small and randomly distributed, and they are also interfered with by complex backgrounds. Some ROIs may no longer be salient to the observer, so they are challenging to be recognized by single-image saliency proposals. Cutting large-scale images into small tiles can better make saliency detection work.

E. Limitations

The proposed framework is based on fusion, which obtains more accurate results by combining the advantages of multiple existing saliency detection methods. However, our fusion strategy is based on a voting mechanism; when most of the adopted saliency proposals only provide very limited or imprecise results, the final performance of the proposed method may be unsatisfactory. Fig. 19 shows an example: although the saliency proposal of DSR is better than others, the fused result is unsatisfactory due to the voting criterion. In addition, due to the high computational complexity of matrix inversion in manifold ranking and energy function optimization in MRF,

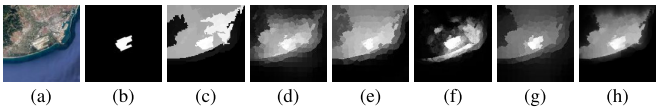


Fig. 19. Challenging example where our model fails. (a) Input image. (b) Ground truth. (c)–(g) Single-image saliency proposals from HS [6], MC [12], GMR [13], DSR [14], and GR [15]. (h) Our fused map.

the computational efficiency of our method needs to be further improved in the future work.

VI. Conclusion

This article proposes an effective and unsupervised cosaliency detection and ROI extraction framework via manifold ranking and MRF for RSIs. We first introduce a new method for converting SISMs to MISMs via a two-stage manifold ranking by considering the correlation information among the image set. Then, we present a novel two-scale MRF-based saliency fusion method, which takes both local and global cues of the images into account. The benefit of converting SISMs to MISMs is that it well suppresses the regions that are salient in a single image but not repeated appearance in the image group. Moreover, the locally adaptive fusion strategy for complementary MISMs generates more accurate cosaliency maps. In particular, we adopt simple and efficient single-image saliency detection approaches as proposals, which reduces the complexity of our model. Evaluation on the GeoEye-1, SPOT-5, Google Earth-Residential, and Google Earth-Airport datasets has demonstrated that our proposed method outperforms some state-of-the-art methods in both qualitative and quantitative results. In future work, we plan to apply deep learning techniques for more powerful feature extraction. In addition, in the fusion strategy, how to better evaluate saliency proposals to guide the fusion process is a more interesting and challenging topic. Finally, a more efficient saliency proposal fusion strategy will be considered in future research.

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